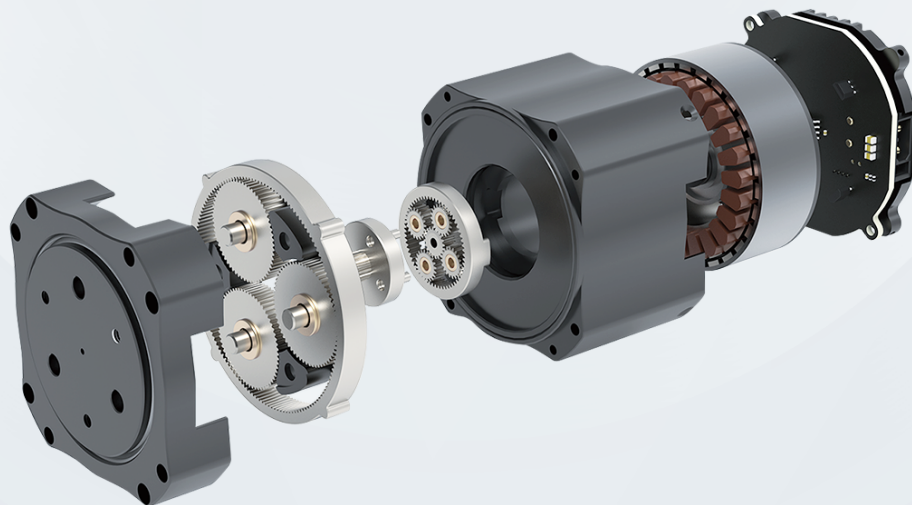


JOINT MODULE PRODUCT MANUAL | SERVO JOINT MODULE
OF HIGH TORQUE

关节模组产品手册

高擎动力伺服关节模组



HTDW-4438-30-NE

(HTDW-4530-02-DNE)

行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	44
高度 Height (mm)	43.4
重量 weight (g)	237

模组性能参数

减速比 Reduction ratio	30
额定电压 Rated Voltage (V)	24
额定电流 Rated Current (A)	0.7
额定扭矩 Rated torque(Nm)	2

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

Module Performance Parameters

堵转扭矩 Locked-rotor torque(Nm)	10
额定转速 Rated RPM(RPM)	40
空载转速 No-load RPM(RPM)	160
额定输出功率 Rated Output Power(W)	8.5

模组电气性能参数

电压范围 Voltage Range(V)	12-50.4
峰值电流 Peak current(A)	10
转速控制精度 Speed control accuracy(Nm/A)	±10%
输入端编码器分辨率 High-speed encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5

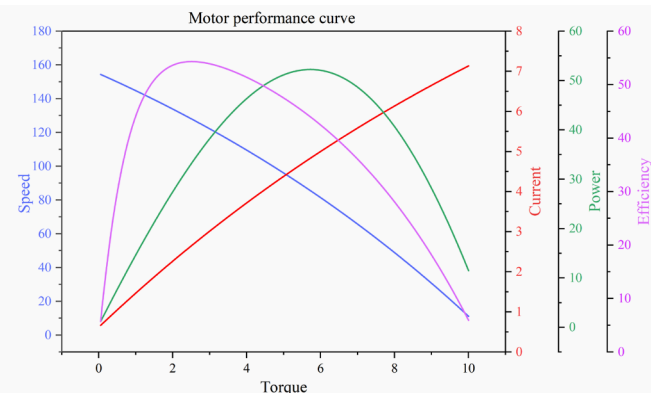
Electrical Performance Parameters of the Module

通讯协议 Communication	CANFD/CAN
扭矩控制精度 control accuracy(Nm/A)	±10%
响应时间 Response time(us)	140
输出端编码器分辨率 Low-speed encoder resolution(bit)	14
控制频率 Control frequency(Hz)	3K

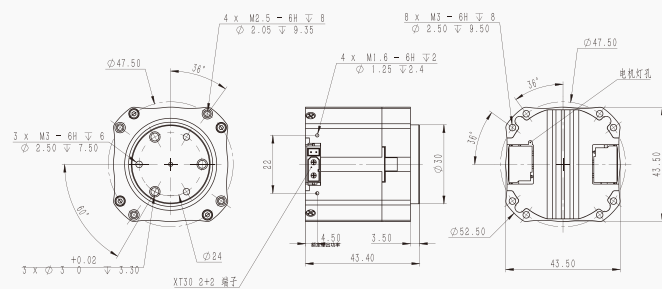
电机特性参数

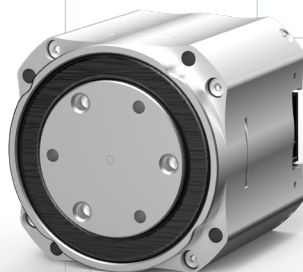
扭矩常数 Torque constant(Nm/A)	0.272	极对数 Polar logarithm	14
-------------------------------	-------	------------------------	----

TN曲线 TN curve



产品安装尺寸图纸 Product Installation Dimension Drawing





HTDW-4438-30-NE-JC

(交叉滚子轴承版本) 行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	44
高度 Height (mm)	44.9
重量 weight (g)	237

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.272	极对数 Polar logarithm	14
-------------------------------	-------	------------------------	----

模组性能参数

减速比 Reduction ratio	30
额定电压 Rated Voltage (V)	24
额定电流 Rated Current (A)	0.7
额定扭矩 Rated torque(Nm)	2

Module Performance Parameters

堵转扭矩 Locked-rotor torque(Nm)	10
额定转速 Rated RPM(RPM)	40
空载转速 No-load RPM(RPM)	160
额定输出功率 Rated Output Power(W)	8.5

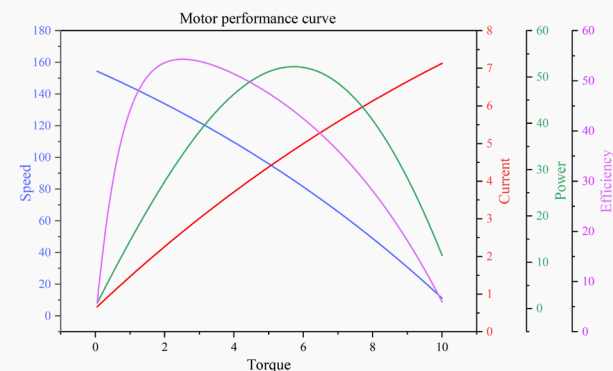
模组电气性能参数

电压范围 Voltage Range(V)	12-50.4
峰值电流 Peak current(A)	10
转速控制精度 Speed control accuracy(Nm/A)	±10%
输入端编码器分辨率 High-speed encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5

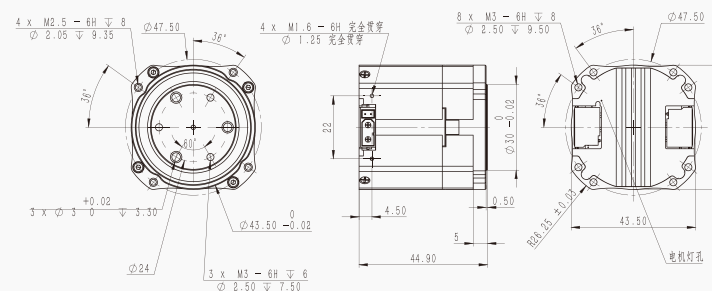
Electrical Performance Parameters of the Module

通讯协议 Communication	CANFD/CAN
扭矩控制精度 control accuracy(Nm/A)	±10%
响应时间 Response time(us)	140
输出端编码器分辨率 Low-speed encoder resolution(bit)	14
控制频率 Control frequency(Hz)	3K

TN曲线 TN curve



产品安装尺寸图纸 Product Installation Dimension Drawing



HTDW-5022-02-DNE

行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	50
高度 Height (mm)	47.4
重量 weight (g)	322

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.54	极对数 Polar logarithm	14
-------------------------------	------	------------------------	----

模组性能参数

减速比 Reduction ratio	22
额定电压 Rated Voltage (V)	24
额定电流 Rated Current (A)	2
额定扭矩 Rated torque(Nm)	3.5

Module Performance Parameters

堵转扭矩 Locked-rotor torque(Nm)	13
额定转速 Rated RPM(RPM)	82
空载转速 No-load RPM(RPM)	123
额定输出功率 Rated Output Power(W)	30

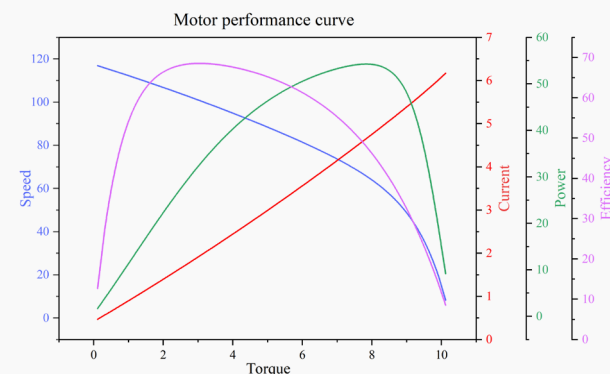
模组电气性能参数

电压范围 Voltage Range(V)	12-50.4
峰值电流 Peak current(A)	10
转速控制精度 Speed control accuracy(Nm/A)	±10%
输入端编码器分辨率 High-speed encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5

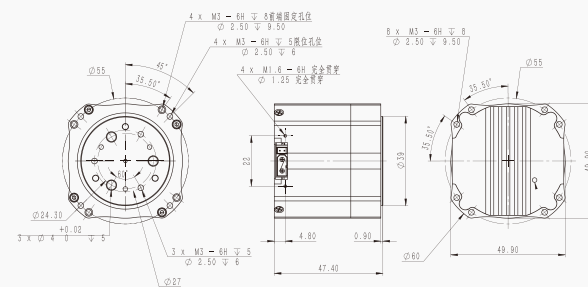
Electrical Performance Parameters of the Module

通讯协议 Communication	CANFD/CAN
扭矩控制精度 control accuracy(Nm/A)	±10%
响应时间 Response time(us)	140
输出端编码器分辨率 Low-speed encoder resolution(bit)	14
控制频率 Control frequency(Hz)	3K

TN曲线 TN curve



产品安装尺寸图纸 Product Installation Dimension Drawing



HTDW-5036-02-DNE

行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	50
高度 Height (mm)	47.4
重量 weight (g)	323

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.54	极对数 Polar logarithm	14
-------------------------------	------	------------------------	----

模组性能参数

减速比 Reduction ratio	36
额定电压 Rated Voltage (V)	24
额定电流 Rated Current (A)	2
额定扭矩 Rated torque(Nm)	6

Module Performance Parameters

堵转扭矩 Locked-rotor torque(Nm)	21
额定转速 Rated RPM(RPM)	50
空载转速 No-load RPM(RPM)	75
额定输出功率 Rated Output Power(W)	30

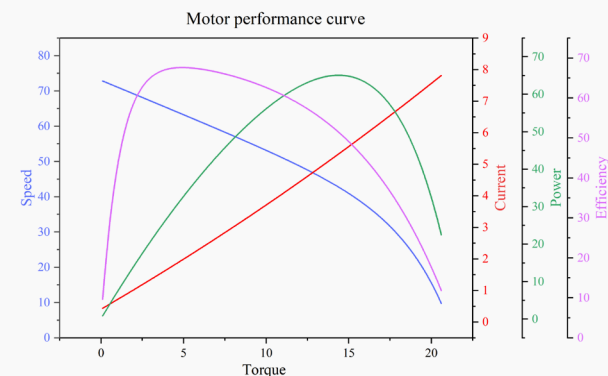
模组电气性能参数

电压范围 Voltage Range(V)	12-50.4
峰值电流 Peak current(A)	10
转速控制精度 Speed control accuracy(Nm/A)	±10%
输入端编码器分辨率 High-speed encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5

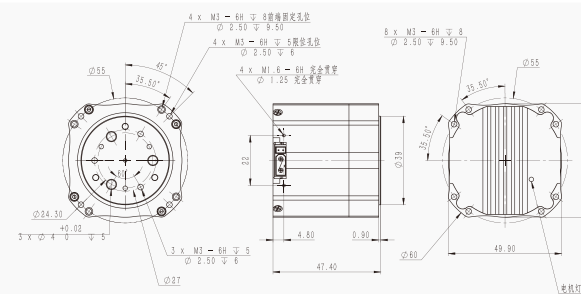
Electrical Performance Parameters of the Module

通讯协议 Communication	CANFD/CAN
扭矩控制精度 control accuracy(Nm/A)	±10%
响应时间 Response time(us)	140
输出端编码器分辨率 Low-speed encoder resolution(bit)	14
控制频率 Control frequency(Hz)	3K

TN曲线 TN curve



产品安装尺寸图纸 Product Installation Dimension Drawing



High Torque

HTDW-5036-02-CNE

(交叉滚子轴承版本) 行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	50
高度 Height (mm)	47.4
重量 weight (g)	345

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.54	极对数 Polar logarithm	14
-------------------------------	------	------------------------	----

模组性能参数

减速比 Reduction ratio	36
额定电压 Rated Voltage (V)	24
额定电流 Rated Current (A)	2
额定扭矩 Rated torque(Nm)	6

Module Performance Parameters

堵转扭矩 Locked-rotor torque(Nm)	21
额定转速 Rated RPM(RPM)	50
空载转速 No-load RPM(RPM)	75
额定输出功率 Rated Output Power(W)	30

模组电气性能参数

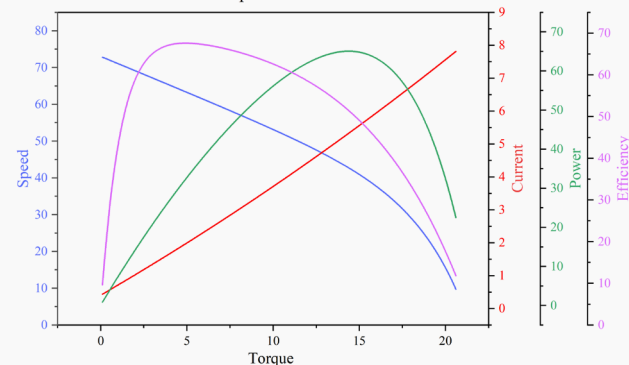
电压范围 Voltage Range(V)	12-50.4
峰值电流 Peak current(A)	10
转速控制精度 Speed control accuracy(Nm/A)	±10%
输入端编码器分辨率 High-speed encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5

Electrical Performance Parameters of the Module

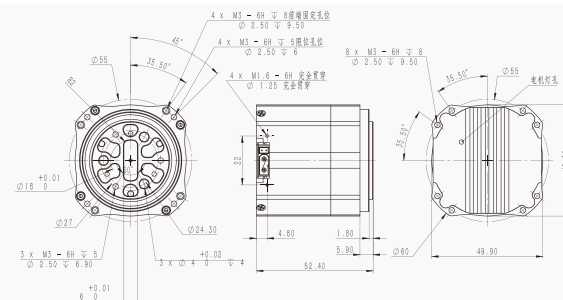
通讯协议 Communication	CANFD/CAN
扭矩控制精度 control accuracy(Nm/A)	±10%
响应时间 Response time(us)	140
输出端编码器分辨率 Low-speed encoder resolution(bit)	14
控制频率 Control frequency(Hz)	3K

TN曲线 TN curve

Motor performance curve



产品安装尺寸图纸 Product Installation Dimension Drawing



基本参数 Basic parameters

减速机特性参数 Reducer Characteristics

模组性能参数

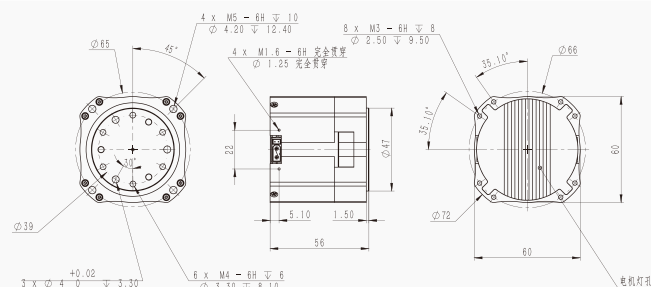
Module Performance Parameters

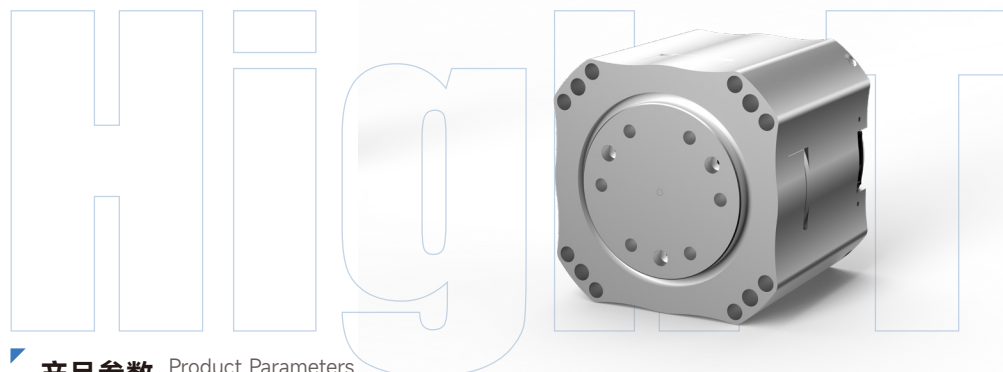
模组电气性能参数

Electrical Performance Parameters of the Module

电机特性参数

TN曲线 TN curve





HTDW-6036-02-CNE

(交叉滚子轴承版本)

行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	60
高度 Height (mm)	56
重量 weight (g)	570

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.531	极对数 Polar logarithm	14
-------------------------------	-------	------------------------	----

模组性能参数 Module Performance Parameters

减速比 Reduction ratio	36
额定电压 Rated Voltage (V)	24
额定电流 Rated Current (A)	3
额定扭矩 Rated torque(Nm)	10

Module Performance Parameters

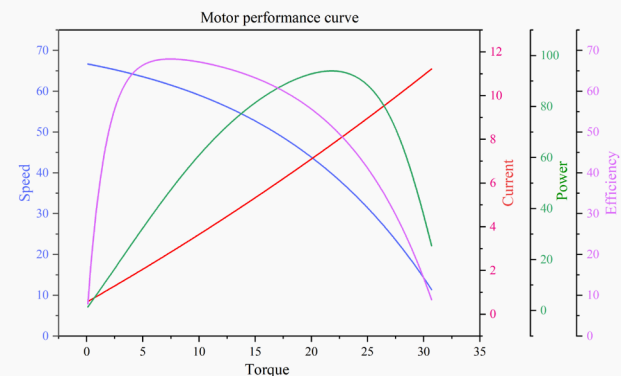
堵转扭矩 Locked-rotor torque(Nm)	36
额定转速 Rated RPM(RPM)	50
空载转速 No-load RPM(RPM)	60
额定输出功率 Rated Output Power(W)	52

模组电气性能参数 Electrical Performance Parameters of the Module

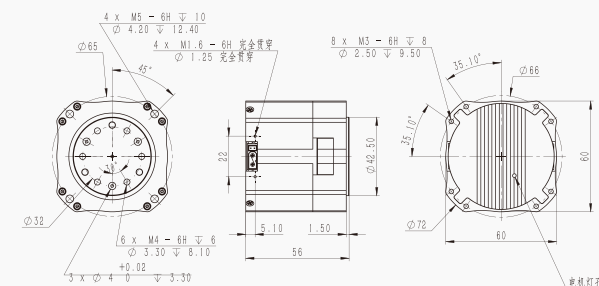
电压范围 Voltage Range(V)	12-50.4
峰值电流 Peak current(A)	15
转速控制精度 Speed control accuracy(Nm/A)	±10%
输入端编码器分辨率 High-speed encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5

通讯协议 Communication	CANFD/CAN
扭矩控制精度 control accuracy(Nm/A)	±10%
响应时间 Response time(us)	140
输出端编码器分辨率 Low-speed encoder resolution(bit)	14
控制频率 Control frequency(Hz)	3K

TN曲线 TN curve



产品安装尺寸图纸 Product Installation Dimension Drawing



HTPW-7507-02-DNE

行星关节模组

产品参数 Product Parameters

基本参数 Basic parameters

长宽 length and width (mm)	75
高度 Height (mm)	35
重量 weight (g)	390

减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

电机特性参数 Motor parameters

扭矩常数 Torque constant(Nm/A)	0.602	极对数 Polar logarithm	14
-------------------------------	-------	------------------------	----

模组性能参数

减速比 Reduction ratio	7
额定电压 Rated Voltage (V)	24
额定电流 Rated Current (A)	5.6
额定扭矩 Rated torque(Nm)	4

Module Performance Parameters

堵转扭矩 Locked-rotor torque(Nm)	12
额定转速 Rated RPM(RPM)	250
空载转速 No-load RPM(RPM)	270
额定输出功率 Rated Output Power(W)	92

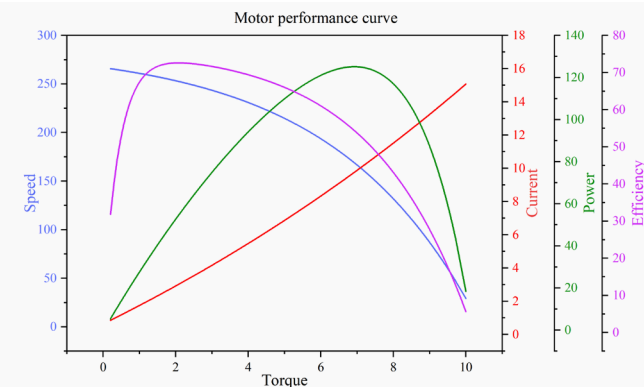
模组电气性能参数

电压范围 Voltage Range(V)	12-50.4
峰值电流 Peak current(A)	20
转速控制精度 Speed control accuracy(Nm/A)	±10%
输入端编码器分辨率 High-speed encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5

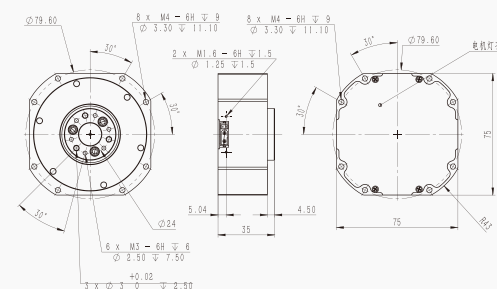
Electrical Performance Parameters of the Module

通讯协议 Communication	CANFD/CAN
扭矩控制精度 control accuracy(Nm/A)	±10%
响应时间 Response time(us)	140
输出端编码器分辨率 Low-speed encoder resolution(bit)	14
控制频率 Control frequency(Hz)	3K

TN曲线 TN curve



产品安装尺寸图纸 Product Installation Dimension Drawing



HTDW-7535-02-CNE

(交叉滚子轴承版本) 行星关节模组

产品参数 Product Parameters

基本参数

Basic parameters

长宽 length and width (mm)	75
高度 Height (mm)	56
重量 weight (g)	850

减速机特性参数

Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

电机特性参数

Motor parameters

扭矩常数 Torque constant(Nm/A)	0.602	极对数 Polar logarithm	14
-------------------------------	-------	------------------------	----

模组性能参数

Module Performance Parameters

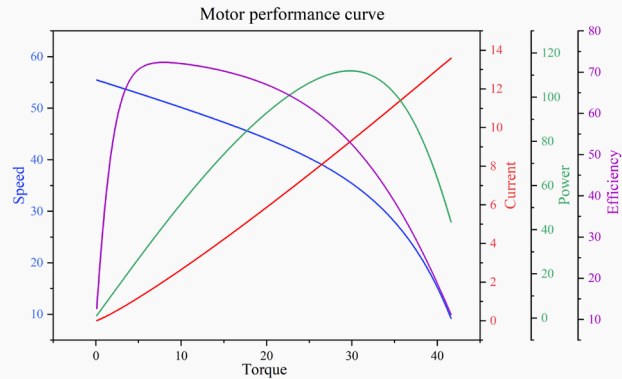
减速比 Reduction ratio	35	堵转扭矩 Locked-rotor torque(Nm)	60
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	50
额定电流 Rated Current (A)	4.2	空载转速 No-load RPM(RPM)	60
额定扭矩 Rated torque(Nm)	18	额定输出功率 Rated Output Power(W)	94

模组电气性能参数

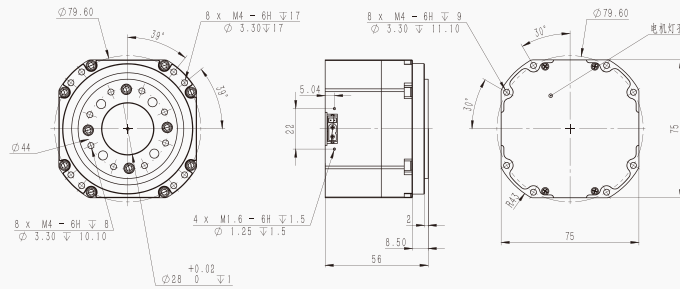
Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	20	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speed encoder resolution(bit)	14	输出端编码器分辨率 Low-speed encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

TN曲线 TN curve



产品安装尺寸图纸 Product Installation Dimension Drawing



HTPU-7033-04-CNE

超薄系列关节模组

产品参数 Product Parameters

基本参数

Basic parameters

长宽 length and width (mm)	140.5X71.5
高度 Height (mm)	36.9
重量 weight (g)	808

减速机特性参数

Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

电机特性参数

Motor parameters

扭矩常数 Torque constant(Nm/A)	0.59	极对数 Polar logarithm	14
-------------------------------	------	------------------------	----

模组性能参数

Module Performance Parameters

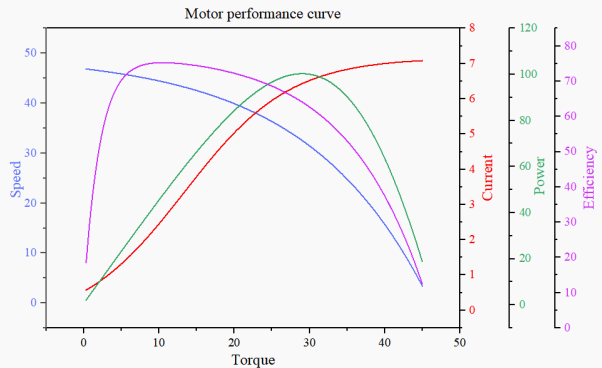
减速比 Reduction ratio	33	堵转扭矩 Locked-rotor torque(Nm)	50
额定电压 Rated Voltage (V)	24	额定转速 Rated RPM(RPM)	40
额定电流 Rated Current (A)	4.8	空载转速 No-load RPM(RPM)	48
额定扭矩 Rated torque(Nm)	20	额定输出功率 Rated Output Power(W)	84

模组电气性能参数

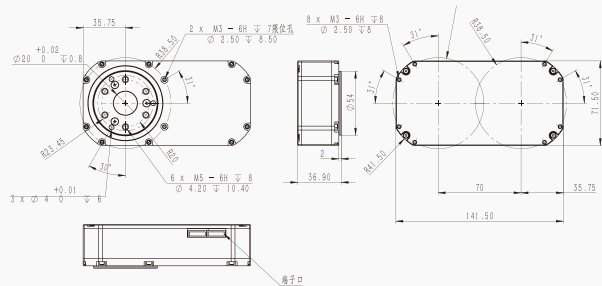
Electrical Performance Parameters of the Module

电压范围 Voltage Range(V)	12-50.4	通讯协议 Communication	CANFD/CAN
峰值电流 Peak current(A)	20	扭矩控制精度 control accuracy(Nm/A)	±10%
转速控制精度 Speed control accuracy(Nm/A)	±10%	响应时间 Response time(us)	140
输入端编码器分辨率 High-speed encoder resolution(bit)	14	输出端编码器分辨率 Low-speed encoder resolution(bit)	14
通讯波特率 Baud rate(Mbps)	5	控制频率 Control frequency(Hz)	3K

TN曲线 TN curve



产品安装尺寸图纸 Product Installation Dimension Drawing



产品参数 Product Parameters

基本参数 Basic parameters

长宽	44
length and width (mm)	
高度	44.9
Height (mm)	
重量	237
weight (g)	

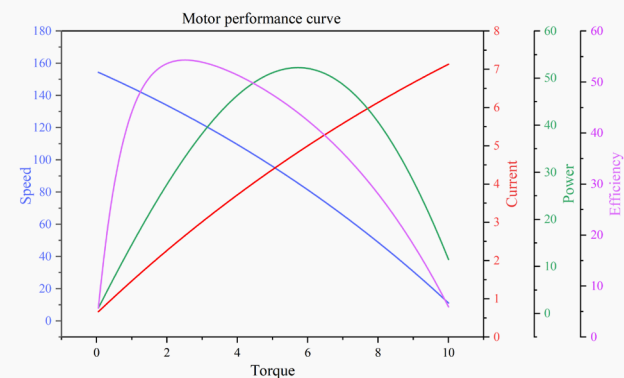
减速机特性参数 Reducer Characteristics

齿隙 Backlash(Arcmin)	<10
重复定位精度 Repeat Positioning Accuracy(Arcmin)	<12
绝对定位精度 Absolute positioning accuracy(Arcmin)	<15

电机特性参数

扭矩常数 Torque constant(Nm/A)	0.272	极对数 Polar logarithm	14
-------------------------------	-------	------------------------	----

TN曲线 TN curve



模组性能参数

减速比 Reduction ratio	31
额定电压 Rated Voltage (V)	24
额定电流 Rated Current (A)	0.7
额定扭矩 Rated torque(Nm)	2

Module Performance Parameters

堵转扭矩 Locked-rotor torque(Nm)	10
额定转速 Rated RPM(RPM)	40
空载转速 No-load RPM(RPM)	160
额定输出功率 Rated Output Power(W)	8.5

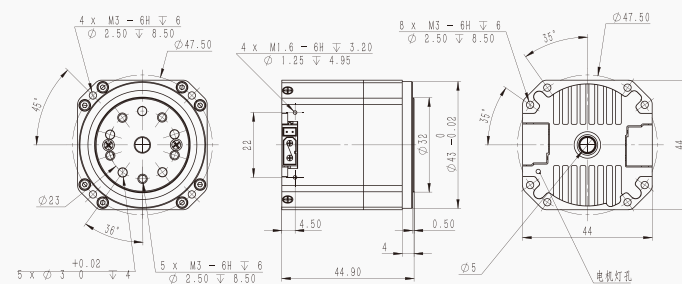
模组电气性能参数

电压范围	12-50.4
Voltage Range(V)	
峰值电流	10
Peak current(A)	
转速控制精度	±10%
Speed control accuracy(Nm/A)	
输入端编码器分辨率	14
High-speeded encoder resolution(bit)	
通讯波特率	5
Baud rate(Mbps)	

Electrical Performance Parameters of the Module

通讯协议	CANFD/CAN
Communication	
扭矩控制精度	±10%
control accuracy(Nm/A)	
响应时间	140
Response time(us)	
输出端编码器分辨率	14
Low-speeded encoder resolution(bit)	
控制频率	3K
Control frequency(Hz)	

产品安装尺寸图纸 Product Installation Dimension Drawing



基本参数 Basic parameters

减速机特性参数 Reducer Characteristics

模组性能参数

Module Performance Parameters

模组电气性能参数

Electrical Performance Parameters of the Module

电机特性参数

TN曲线 TN curve

